📘 **Mobile Computing Cheat Sheet**

Lesson 1 – Basics

- Mobile computing = using tech while moving (phones, laptops, tablets).
- Types: wireless, nomadic, ubiquitous, pervasive.
- **Wireless vs Mobile**:
    - Wireless = no cables.
    - Mobile = adds movement + roaming.
- **Cellular & PCS**: both use radio frequencies for phone/data.
- **Limitations**: slow speeds, security risks, battery drain, interference, health concerns.
- **Applications**: vehicles (traffic updates), emergencies, salesmen (mobile office), web access, entertainment.
- **Benefits**: cheaper, faster setup, portable, flexible.

Lesson 2 – Wireless Technologies

- Types of networks: LAN, WLAN, WAN, MAN, PAN, CAN, SAN.
- **PAN**: Bluetooth, Zigbee → short range (headsets, car systems).
- **WLAN**: Wi-Fi → local internet access.
- **MAN/WiMAX**: city-wide coverage, faster than Wi-Fi.
- **Issues**: interoperability, cost, limited range.

Lesson 3 – Middleware & Standards

- Middleware = software that helps apps work smoothly over wireless.
- Functions: queues messages, reduces data, hides disconnections.
- **IEEE 802.11 (Wi-Fi)** standards:
    - b = slower, cheap.
    - a/g = faster.
    - n = very fast, better range.
- Bluetooth = short range, low power.

Lesson 4 – Wireless Networks & Routing

- **Ad-hoc networks (MANETs)** = devices connect directly, no fixed infrastructure.
- Routing types:
    - Proactive (tables), Reactive (on demand), Hybrid (mix).
    - Flow-oriented, Adaptive, Hierarchical, Geographical, Power-aware.
- Intelligent agents = software that acts for you (privacy, data mining, apps).

Lesson 5 – Mobile Communications

- Cellular = divided into **cells** with base stations.
- **Frequency reuse** = same frequencies reused in different cells.
- **Call process**: control channels set up → voice channels carry call.
- **Handoff** = switching between cells when moving.
- **Propagation effects/issues**: fading, interference, signal strength issues.

Lesson 6 – Data Transmission

- **Generations**:
    - 1G = analog.
    - 2G = digital (GSM, CDMA).
    - 3G = faster, global roaming.
- GSM = most common worldwide, uses TDMA.
- Multiple access methods: FDMA, TDMA, CDMA, SDMA.
- Mobile internet tech: GSM, GPRS, EDGE, VOIP.

Lesson 7 – Mobile IP

- Mobile IP = keeps your internet connection while moving.
- Two addresses:
    - Home address (permanent).
    - Care-of address (temporary).
- Steps: Agent discovery → Registration → In service.

- TCP/IP layers: Application, Transport, Network, Link.
- Mobile vs Nomadic:
  - Mobile = seamless connection.
  - Nomadic = reconnect each time.

Lesson 8 – Internet Protocols

- **IPv4**: 32-bit addresses (limited, insecure, no auto-config).
- **IPv6**: 128-bit addresses (huge space, built-in security, auto-config).
- Address types: Unicast (one-to-one), Broadcast (everyone), Multicast (group).
- IPv6 advantages: route optimization, no foreign agents, better security.

Lesson 9 – Mobile Databases

- Mobile DB = database accessed via mobile device.
- Works offline → syncs later.
- Needs: interactivity, local hardware access, bandwidth saving.
- Architecture: fixed hosts, mobile units, base stations.
- Vendors: Sybase SQL Anywhere, IBM DB2, Microsoft SQL Compact, Oracle Lite.
- Used in retail, logistics, aviation, transport.

Lesson 10 – Wireless Sensor Networks (WSN)

- WSN = many small sensors communicating wirelessly (temperature, motion, pollution).
- Components: processor, memory, sensors, radio, power.
- Characteristics: low power, fault tolerance, dynamic topology.
- Challenges: scalability, mobility, security, QoS.
- Applications: military, healthcare, environment, smart grids, surveillance, disaster response.

🎯 Quick Takeaway

Mobile computing = **tech on the move**. It relies on **wireless networks, standards (Wi-Fi, Bluetooth, GSM), smart routing, and databases**. Challenges = **bandwidth, battery, security, interference**. Future = **IPv6, sensor networks, smarter mobile IP**.